HICHAM TBINA

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Power Electronics / Drives Engineer — Converters, Inverters & Traction Systems

PROFESSIONAL SUMMARY

Electrical Engineering specializing in high-performance power conversion and traction systems. Experienced in designing and validating hardware-in-the-loop systems, including traction inverters and Intelligent BESS. Delivered X kW hardware solutions with significant efficiency improvements through advanced control (MPC, Nonlinear) and Reliability Analysis (WCA). Open to national and international opportunities .

TECHNICAL SKILLS

- **Power Electronics:** Converter Topologies (DC-DC, Zeta, Buck/Boost), Inverters, PWM/MLI, Power MOSFET/IGBT Selection, EMI Mitigation, Thermal Design.
- **Control & Simulation:** MATLAB/Simulink (Simscape), PLECS, LTspice, Advanced Control (PID, MPC, State-Space, Nonlinear Backstepping).
- **Energy Systems:** Battery Management Systems (BMS), SoC/SoH Estimation, Grid-forming Control, V2G Integration, Renewable Energy Coordination.
- **Industrial & Embedded:** PLCnext, CODESYS, Modbus TCP/IP, HMI Development, Altium Designer (PCB Layout), Hardware-in-the-Loop (HIL) testing.

PROFESSIONAL EXPERIENCE

Vulcain Engineering (Consultant for TCMS) — Feb 2026 – Present
Traction System MVS Intern — *Fes, Morocco*

- **Medium Voltage Supply (MVS):** Analyzing and designing Medium Voltage Supply functions (FPS, ADL1, SWDS) for rail traction systems using control algorithms and schematics.
- **Control Logic:** Modeling power flow and control sequences in MATLAB Stateflow; integrating via OPC UA with CODESYS for real-time HMI supervision.

Lear RD — July 2025 – Sept 2025
Hardware RD Intern — *Rabat, Morocco*

- **Reliability WCA:** Performed Worst-Case Analysis (WCA) on ideal diode circuits to ensure high-reliability performance in automotive power modules.
- **Automation:** Developed VBA scripts to automate simulation data processing, reducing analysis time for embedded electronic reliability tests.

Redal-BCC — July 2024 – Sept 2024
SCADA Systems Intern — *Rabat, Morocco*

- **Network Modernization:** Migrated SCADA Power CC to LYNX systems to modernize grid supervision and improve load management efficiency.

PROJECTS & RESEARCH

- **Intelligent BESS with Adaptive Virtual Inertia:** Developed an AI-based storage system using Reinforcement Learning to stabilize grids with high renewable penetration and provide grid-forming control.
- **UPS Design And SMPS:** Engineered a switched-mode power supply featuring a lithium BMS and a SPWM-controlled inverter for stable AC output.
- **PV System with Nonlinear Backstepping Control:** Designed a high-efficiency PV system utilizing a Zeta-based bidirectional converter with MPPT and coordinated CC/CV battery management[cite: 6, 7].
- **Li-ion Battery State Observer:** Developed a state-space observer for joint SoC and core temperature estimation, enabling adaptive fast-charging protocols in Simulink.
- **Variable Voltage DC/DC Regulator:** Implemented a PID-controlled DC/DC converter optimizing transient response and stable output under varying loads.

EDUCATION

ENSET Mohammedia — 2023 – 2026
M.S. Electrical Engineering (Industrial Control) — *Mohammedia, Morocco*

BTS Salé — 2021 – 2023
Higher Technician Certificate in Electrotechnics — *Salé, Morocco*

LANGUAGES

- **English:** Fluent (Professional/Technical Proficiency).
- **French:** Technical, Written, and Oral Proficiency.
- **Arabic:** Native.